

FaceLinkGen: Rethinking Identity Leakage in Privacy-Preserving Face Recognition

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Abstract

Frequency-domain transformation-based privacy-preserving face recognition (PPFR) systems protect user identity by converting raw face images into protected templates in the frequency domain before recognition. Existing evaluations treat privacy as resistance to pixel-level reconstruction, measuring attack success with PSNR and SSIM. We show that this evaluation paradigm may not adequately capture identity-level privacy: preventing pixel-level reconstruction does not necessarily prevent identity leakage. We present FaceLinkGen, an identity-centric attack that performs identity linkage and face regeneration directly from protected templates without recovering original pixels. Treating the conversion process as a black-box oracle, FaceLinkGen achieves over 98.5% matching accuracy and above 96% regeneration success on three recent frequency-domain PPFR systems. These results reveal a gap between pixel-distortion metrics and identity-level privacy in the evaluated methods, and motivate complementing pixel-level metrics with identity-centric evaluation in frequency-domain PPFR research.

CCS Concepts

• Security and privacy → Biometrics; • Computing methodologies → Computer vision.

Keywords

privacy-preserving face recognition, identity leakage, face regeneration, linkage attack

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1 Introduction and Related Works

The fundamental promise of transformation-based Privacy-Preserving Face Recognition (PPFR) systems is compelling: verify a user's identity without ever exposing their raw facial data [6, 14, 19–21, 37]. A user's face is converted into a protected template that a recognition server can use for matching, but that, ideally, cannot be reversed to recover the original image or used to infer private attributes. The primary adversary in this setting is the curious or malicious service provider who receives the template [8, 13].

The prevailing evaluation paradigm for these systems is currently significantly limited. Historically, the dominant objective has been to prevent the reconstruction of the original registration image, with attack success measured through pixel-level or local similarity metrics such as peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM), a legacy inherited from

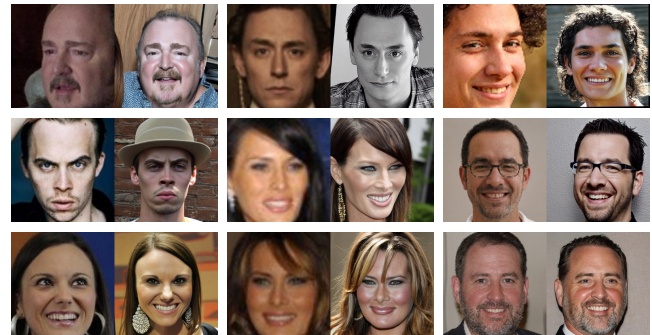


Figure 1: Regeneration attack results. In each subplot, the left is the original image and the right is the regenerated image from the protected template. Each row shows examples from one PPFR method, in the order of PartialFace, MinusFace, and FracFace.

image privacy and compression literature. While some recent approaches, such as FracFace [6], also report identity similarity and employ StyleGAN for non-pixel-level reconstruction, they still have a strong reconstruction-centric focus. A large body of prior work adopts this paradigm, using resistance to pixel-level recovery as evidence of privacy protection and optimizing attack objectives accordingly. Representative systems, including DuetFace [19], PartialFace [20], MinusFace [21], FaceObfuscator [14], and FracFace, explicitly rely on this reconstruction-based evaluation to argue robustness against recovery attacks.

This evaluation paradigm, however, rests on a critical implicit assumption: that preventing pixel-level reconstruction is both necessary and sufficient to prevent identity leakage. In this paper, we show that this assumption does not hold. Crucially, compromising privacy does not require recovering the original registered image, nor does pixel-level similarity reliably correspond to identity consistency. In the facial domain, two images that are visually or pixel-wise similar may represent different identities and images that represent the same identity may share no pixel or local similarity. CanFG [35] can generate two images with very high pixel-level similarity yet in completely different identities; conversely, in daily life, any two arbitrary photos of the same person (one ID photo and one social media photo) would have very high identity similarity yet likely very low pixel-level or structure-level similarity. An example is provided in Figure 2.

This misconception misled not only the evaluation but also the simulated attack design because of the overlooked fact that identity-revealing information can remain accessible even when pixel-level reconstruction is infeasible. By employing pixel-level loss functions, simulated attackers (red-team researchers) are inadvertently trapped into pursuing the specific registration image as ground truth. This objective is often mathematically impossible

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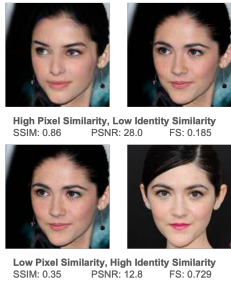


Figure 2: SSIM and PSNR are not always correlated with identity correlation.



Figure 3: Using pixel-level loss or StyleGAN will yield unsuccessful reconstruction compared to our ID-guided method.

Table 1: Comparison between pixel-level metrics and identity-level metrics on two cases: a protected face generated by CanFG with its original face, and two images of the same person. We can see that higher pixel-level similarity does not mean higher identity-level similarity.

Compared With	SSIM	PSNR	MSE	FS
Another Face	0.235	10.44	6699	0.586
CanFG Face	0.841	26.81	143	0.008

due to the information loss in protection, causing the generator to produce a blurry image, likely an average of all images in the dataset. The failure is illustrated in Figure 3. Although FracFace [6] also reports identity similarity, the simulated U-Net attack remains reconstruction-based, aiming to recover the original pixels. The StyleGAN [16] attack also failed to generate a face with the same identity as the original image, as shown in Figure 3.

Recent work has begun to challenge the reconstruction-centric perspective in non-facial domains by shifting attention toward semantic-level inversion. In such settings, the attacker aims not to recover the original image itself, but to regenerate information that is semantically consistent with the original identity. This goal is both easier to achieve and more aligned with realistic attack objectives. For instance, Yue et al. [42] proposes a semantic recovery framework that leverages generative models to synthesize semantically consistent images without pixel-level similarity in federated learning attacks. Complementary findings further suggest that traditional reconstruction metrics fail to capture how humans perceive privacy leakage [30].

Our approach differs from standard Model Inversion Attacks (MIAs). One category of MIA, represented by Wang et al. [33], recovers original images from embeddings. For many deep facial embedding models, such processes resemble image generation tasks rather than adversarial attacks, as they exploit the inherent invertibility of learned representations. Similar works exist in the ID-controlled image generation domain, like Arc2Face [25], PuLID [11], and FaceID IP-Adapter [39]. In contrast, our method targets

structural vulnerabilities in the template generation process itself. Since this conversion often utilizes rule-based transformations independent of specific deep models, the attack surface differs from embedding-based reconstruction. Another category of MIA, which is closer to the original definition [9], such as the ones prevented by Wang et al. [36], aims to reconstruct the training dataset to compromise identity privacy in the *training set*. This focus on training data deviates from our objective of protecting individual user templates. Existing solutions for training-level privacy include FaceMAE [34] and the use of synthetic data [4].

Our Contribution. This paper argues that the prevailing evaluation paradigm for frequency-domain PPFR, which measures privacy as resistance to pixel-level reconstruction, may not adequately capture identity-level leakage. We introduce and validate an identity-centric evaluation approach, and present FaceLinkGen, an attack that performs identity linkage and face regeneration directly from protected templates without recovering original pixels. We show that pixel-level metrics can overestimate the privacy provided by the evaluated frequency-domain PPFR systems: identity information can be extracted via a straightforward distillation pipeline even when the original image remains “unreconstructable” under PSNR and SSIM. We additionally provide pilot evaluations on an adversarial de-identification method (TIP-IM) and a non-frequency-domain method (CanFG), with preliminary results suggesting that similar vulnerabilities may extend beyond the frequency-domain PPFR family. The simplicity of the pipeline is intentional: if a standard procedure suffices, the vulnerability is in the representation. This work encourages identity-centric evaluation alongside existing metrics in future PPFR and de-identification research.

2 Threat Model

The PPFR paradigm was originally motivated by a specific and well-defined concern: a user must submit face data to a remote recognition server they do not fully trust [8]. Since the raw face image is deeply personal, the user has reason to want the server to perform recognition without gaining access to the original biometric. PPFR addresses this by transforming the face into a protected representation before transmission, one that the server can use for matching but, ideally, cannot revert to recover the original face, infer soft biometrics such as age, gender, or ethnicity, or link the identity to records outside the service. The primary adversary in this setting is therefore the *curious or malicious service provider*, an insider who legitimately receives the protected template and seeks to exploit it.

Protection against network-level eavesdroppers is a separate concern and was not the original intent of PPFR. Interception of data in transit is well addressed by general-purpose secure communication protocols such as TLS, which are application-agnostic and do not require altering the face representation itself. Nevertheless, some recent PPFR work has adopted an external-attacker framing: Mi et al. [21] describes this attacker as “typically envisioned as a malicious third-party wiretapping the transmission,” departing from the original insider-centric design intent. One might argue that the conversion process and the recognition service could be controlled by separate entities, with the conversion model running locally or under a trusted third party server, thereby preventing the service

provider from accessing the conversion process. However, a locally running model can be reverse-engineered by the service provider to obtain oracle access, and the provider can submit arbitrary faces through the local client to collect paired (original, protected) data on their server. If instead the conversion process runs on the third-party remote server, the trust problem is merely shifted rather than solved: the user must now trust the third-party server with their raw face data, which contradicts the original motivation for PPFR.

We return to first principles and evaluate PPFR against its intended adversary: a service provider with oracle access to the conversion process (the ability to query it with arbitrary inputs and observe outputs), but no knowledge of its internal architecture or parameters. Even under this insider-focused framing, our attacker model assumes less than prior work: Mi et al. [21] assumes knowledge of the conversion architecture (but not the random channel selection parameters), and Mi et al. [20] assumes access to the conversion process but not the selected channel IDs. We assume no knowledge of the architecture, parameters, or hyperparameters. CanFG [35] places all of its security in the secrecy of the conversion model, which is an unrealistic assumption when the service provider itself is considered the adversary. They themselves pointed out that if the paired face data can be obtained, a reverse CanFG model could be trained to reconstruct the original faces.

For completeness, we also note that Zhang et al. [43] requires approximately 6,900 online verification queries per identity and depends on the server returning continuous similarity scores for optimization, a setup highly susceptible to rate limiting and fraud detection. Many deployed FR systems return only binary accept/reject decisions or quantized similarity scores, rendering such approaches infeasible [22, 31]. Our method does not rely on server-side behavior at all.

3 Methods

The simplicity of our method is intentional. We show that even strong protection methods fail with a simple, standard distillation process, proving that the vulnerability resides in the representation itself. To formulate this, we consider a face image X as a combination of identity information z_I and non-identity (nuisance) information z_N , such that $X \sim p(\cdot \mid z_I, z_N)$.

In transformation-based PPFR systems, a protected template T is generated to hide the visual data of X while retaining identity utility. This process can be viewed as a lossy mapping that suppresses the information quantity of z_N while preserving z_I :

$$T \sim p(\cdot \mid z_I). \quad (1)$$

Existing evaluations often equate privacy with the failure of pixel-level reconstruction. However, since z_N is largely discarded, recovering the original pixels X is a severely ill-posed problem. Conventional attacks fail because they attempt to optimize for specific nuisance factors (e.g., exact lighting or pose) that no longer exist in T , resulting in blurry or identity-inconsistent outputs.

Our approach, FaceLinkGen, instead focuses on extracting the remaining identity information. We use a distillation-style procedure to align the template domain with a standard identity embedding space. Given a public dataset, we train a student model f_s to recover an identity representation z'_I from T . The training objective is to maximize the cosine similarity between the student's output and

the embeddings z_t extracted by a frozen teacher model f_t from original images X :

$$\mathcal{L} = 1 - \frac{1}{N} \sum_{k=1}^N s(f_s(t_k), f_t(i_k)), \quad (2)$$

where t_k and i_k are the protected template and original image derived from the same source image of identity k , and $z'_I = f_s(T)$ is the identity feature recovered by the attacker.

Once z'_I is extracted, the attack bypasses the need for pixel reconstruction by leveraging a diffusion-based generative model g_{diff} . Rather than trying to find the original z_N , we substitute the missing information by sampling from the model's stochastic noise ϵ :

$$\epsilon \sim \mathcal{N}(0, I), \quad Y = g_{\text{diff}}(z'_I, \epsilon). \quad (3)$$

Rather than recovering the discarded nuisance factor z_N , we sample a new z'_N via ϵ , bypassing the ill-posed reconstruction problem ($z_N \neq z'_N$). As long as z_I persists in T , an attacker can recover z'_I and combine it with random ϵ to regenerate an identity-consistent face. This implies that visual distortion of z_N does not prevent identity extraction, as recovering z_N is not a prerequisite for a successful attack.

Note that our attack method is independent of the specific face recognition model employed by the PPFR server. We use ArcFace as the teacher and student network solely because it is a widely-adopted, publicly available facial embedding model, and its compatibility with the Arc2Face generative model simplifies our demonstration. Other embedding models like FaceNet [28] can also be used as long as there is a compatible generative model. The server's actual recognition backbone could be any commercial or proprietary model. Our attack only requires that the protected template retains identity-discriminative features that are learnably aligned with some facial embedding space accessible to the attacker, a condition implicitly assumed by any PPFR system that aims to preserve recognition utility.

4 Attack Vectors

The identity extraction framework described above enables two distinct attack vectors. The linkage attack requires only the extracted embedding z'_I and operates entirely in the embedding space, while the regeneration attack additionally employs the diffusion-based generator g_{diff} to synthesize identity-consistent face images.

4.1 Linkage Attack

A linkage attack aims to associate a real-world identity (e.g., a public face image) with a protected identity, or to link two protected templates belonging to the same individual across different leaked databases. The first case is referred to as face-to-template linkage, while the second is template-to-template linkage. ISO/IEC 24745 explicitly requires resistance against template-to-template search, but does not address face-to-template search. This is likely a utility trade-off for the verification needs of the service provider. This attack vector is similar to an attack vector for hashing: when the input space is known, an attacker can enumerate all possible inputs and map each hashed output back to its original input.

In both attack scenarios, the attacker first obtains a query embedding e_q . This embedding can be extracted using either the student

model f_s or the teacher model f_t , depending on the domain of the query data. The adversary then computes embeddings for all protected templates in the leaked database using f_s and performs a nearest-neighbor search. This process can be written as

$$\arg \max_{t \in T} s(e_q, f_s(t)), \quad (4)$$

where $s(\cdot, \cdot)$ denotes cosine similarity.

4.2 Regeneration Attack

As discussed earlier, reconstructing the original enrollment image is unnecessary and, in many cases, impossible. However, once a universal face embedding (e.g., ArcFace) can be extracted from a protected template, modern face generation models can be leveraged. In this work, we use Arc2Face, which takes a facial embedding as input and generates a face image whose embedding matches the input. This allows us to synthesize a realistic face corresponding to the protected template without reconstructing the original image.

5 Experiments and Results

We selected three frequency-domain PPFR methods with accessible source code: PartialFace [20] (ICCV 2023), MinusFace [21] (CVPR 2024), and FracFace [6] (NeurIPS 2025). These represent the mainstream of currently open-sourced transformation-based PPFR work. For distillation, we used a subset of CASIA-WebFace [40] with around 10K identities and 90K images. Importantly, there is no dataset or architecture overlap between the attacked methods and our student model: PartialFace and FracFace are training-free hard-coded transformations, and MinusFace is trained on MS1Mv2 [10], while our attack model is trained on CASIA-WebFace. The facial embedding model is Antelopev2 with one additional 3x3 Conv2D layer prepended to ensure compatibility with different template formats (channel numbers) if needed. Antelopev2 was chosen because it is the backbone that Arc2Face accepts. The dataset is split into a training and a validation set in an 80-20 ratio. For regeneration testing, we used three datasets: the validation hold-out set of CASIA-WebFace, “this person does not exist” (TPDNE) dataset¹, and Labelled Faces in the Wild (LFW) dataset. The hold-out set is used to test the ability of our method in real images while ensuring no ID duplications. The LFW dataset is used to test the cross-dataset performance of our method with distribution shift, and the TPDNE is used as a synthetic dataset to avoid data cross-contamination from Stable Diffusion 1.5 and Arc2Face training data. Compared to the hold-out set and LFW, photos in the TPDNE dataset are also closer to a headshot, which is what is usually used to create the protected templates.

The distillation process was completed in under two hours on a single NVIDIA A6000 GPU for each of the three evaluated methods, at an estimated cost of approximately USD 0.80 to 1.60. The same process can also be executed on consumer GPUs such as the RTX 4090 or RTX 5090 with comparable wall-clock time; in fact, it is theoretically faster on the RTX 5090 due to its newer architecture. These extreme low costs are deliberate: they serve as a lower-bound analysis demonstrating that current protection mechanisms succumb to a lightweight, generic distillation without requiring complex adversarial optimization.

¹Leonidas/this-person-does-not-exist

Table 2: Regeneration attack results on three PPFR methods across three datasets. Success@5: at least one of five generated images passes Face++ verification. Pass@ τ : per-image pass rate at the Face++ threshold corresponding to false acceptance rate τ .

	Success@5	Pass@1e-5	Pass@1e-4	Pass@1e-3
<i>Dataset: TPDNE</i>				
PartialFace	1.000	0.993	0.996	0.998
MinusFace	0.996	0.936	0.970	0.989
FracFace	0.992	0.904	0.957	0.985
<i>Dataset: CASIA-WebFace Hold-Out</i>				
PartialFace	0.992	0.957	0.970	0.982
MinusFace	0.989	0.930	0.958	0.978
FracFace	0.991	0.920	0.950	0.977
<i>Dataset: LFW</i>				
PartialFace	0.988	0.980	0.983	0.986
MinusFace	0.987	0.974	0.981	0.983
FracFace	0.979	0.943	0.961	0.970

5.1 Minimal-Resource Attack

To stress-test the data requirements, we conducted an experiment on FracFace with only approximately 800 images. With higher weight decay and lower batch size to mitigate overfitting, training completed in under 50 seconds, yet the attack still achieved a 97.0% generation pass rate at FAR 1×10^{-5} and 99.5% linkage accuracy. We further reduced the training set to 256 images, obtaining 98.7% linkage accuracy and 90.5% regeneration success. The consistently high performance across these resource-constrained settings suggests that identity information is not meaningfully disrupted by the transformation, but rather preserved in a form that remains easily extractable.

5.2 Linkage Attack

Since all templates are converted to the standard ArcFace domain, we can not only link between original images and protected templates, but also link between two templates from the same or different protection methods. In any case, we are linking two different images (or templates) of the same person, not a face image and its corresponding template. We used the CASIA-WebFace hold-out set for the linkage attack to ensure no identity overlap between training and testing identities. The hold-out dataset size is 2115.

The closed-set 1-to-N linkage results are in Table 4. The original-image-to-original-image linkage (0.88) establishes a performance upper bound. With the WebFace dataset containing 9.3%-13.0% noise [32], perfect linkage is impossible regardless of method. Our attack achieves linkage success rates consistently above 70%, frequently exceeding 80%, essentially reaching the dataset’s theoretical maximum performance. This confirms that the extracted embeddings function as effective identity descriptors for cross-domain matching. Additionally, the 1-to-1 verification accuracy used in the traditional face recognition benchmark (Table 3) remains near 100%

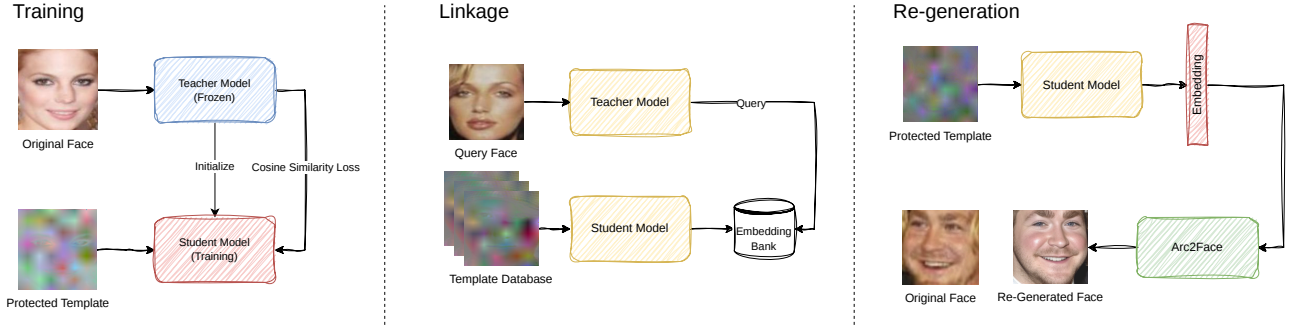


Figure 4: After our model is trained, two main attack vectors can be performed: linkage and re-generation.

Table 3: 1-to-1 verification accuracy between template-to-face and face-to-face on LFW.

	Accuracy	AUROC
MinusFace-to-Face	0.992	0.995
FracFace-to-Face	0.988	0.993
PartialFace-to-Face	0.992	0.996
Face-to-Face	0.998	0.998

Table 4: Linkage Results between MinusFace, PartialFace, FracFace, and Original Image Embeddings on CASIA-WebFace dataset. The query and key for each pair are generated from different images of the same identity. The numbers reported are top-1 recall at a closed set setting.

Query	Key			
	FracFace	MinusFace	PartialFace	Original
FracFace	0.7863	0.7537	0.8137	0.8478
MinusFace	0.7305	0.7206	0.7754	0.8132
PartialFace	0.8028	0.7868	0.8270	0.8572
Original	0.8444	0.8241	0.8563	0.8823

Table 5: Regeneration pass rate cross-verified with the Amazon face comparison API on 700 LFW images.

PartialFace	MinusFace	FracFace
0.99	0.98	0.92

and comparable to the original ArcFace performance, demonstrating that the protection systems fail to meaningfully impede identity matching.

5.3 Regeneration Attack

For the regeneration attack, we evaluate identity recovery on the first 1,000 images from each of the following datasets: the TPDNE dataset, the hold-out set of CASIA-WebFace, and the LFW dataset. Each image is converted into a protected template and mapped to

a facial embedding using the student model. For each embedding, five images are generated using Arc2Face to account for stochasticity. We report both per-image success rate and Success@5. Face generation is highly efficient due to the small backbone of SD1.5: generating a batch of five images for a single embedding takes approximately three seconds on an NVIDIA A6000 GPU, corresponding to a throughput of roughly 1,200 identities per hour and an estimated cost of \$0.0005 per identity generation. Visual examples are shown in Figure 1.

Following the evaluation protocol of CanFG, we employ a commercial face verification system, Face++, to assess identity consistency between the original dataset image and each generated face. We used Face++, marketed to have “financial-grade security standards,” which is usually a higher standard (e.g., more challenging for us) than many open-source methods. This also avoids using the same model (e.g., ArcFace) or models trained on the same datasets (most open source ones) for both embedding extraction and verification. Face++ outputs a confidence score together with three operating thresholds corresponding to error rate of 1×10^{-3} , 1×10^{-4} , and 1×10^{-5} . For each generated image, we record the strictest threshold at which the identity match is accepted and use this as the evaluation outcome. If no face is detected in the original image, it is excluded from the data, while if no face is detected in the generated image, it counts as a failure.

The results are summarized in Table 2. On all three datasets, the success rate at the first attempt is all higher than 97%, and the success rate for five attempts ranges from 97.9% to 100%. Even at the strictest threshold, the success rate is still above 90%. We directly compare our regeneration attack with the original reconstruction attack protocol used by FracFace’s authors. In Table 6, we reported the protection rates claimed in FracFace under its own evaluation and the corresponding rates under our attack on the TPDNE dataset.

The evaluation of attack success in FracFace [6] is based on a Protection (%) metric, defined as the proportion of frequency-domain channels that are filtered or structurally disrupted. This formulation establishes a lower barrier for defensive claims than our identity-centric standard, which requires successful regeneration of images passing commercial-grade verification. By our metric, the protection rate of most recent PPFR methods in U-Net/StyleGAN attack

is almost always 100%. Despite our stricter criterion being unfavorable to reported attack success, we show that high channel protection does not prevent identity leakage: even when FracFace claims high protection under its frequency-domain metric, FaceLinkGen achieves near-total identity recovery. This also shows that channel disruption does not mean identity protection.

To cross-verify this result, we used another commercial facial comparison API from Amazon through EdenAi on 700 selected images on the LFW dataset. The Amazon API only provides a single pass/fail decision with a confidence score; the results are shown in Table 5. The values are close to the Face++ results, validating our claims.

To rule out the dependence on models like Arc2Face or third-party verification services like Face++ or Amazon, we compared the similarity of the extracted embeddings with the original face. As detailed in Section 7, the embedding extracted from a protected template shows higher cosine similarity to its source image than to another image of the same person.

6 Attack Under Constrained Assumptions

Our main results demonstrate that identity information can be reliably extracted by the primary adversary in the PPFR threat model, namely the service provider or developer who has full access to the conversion process [8]. This also follows the threat models in prior PPFR evaluations [6, 20, 21]. We additionally ask how far this vulnerability extends to more constrained adversaries: an external attacker or a knowledge-constrained insider who has neither access to the conversion process nor any knowledge of its internals.

To investigate this, we consider a constrained scenario. In this setting, the attacker possesses only 30 paired image-template samples for validation (not for training) and has limited knowledge of the underlying protection mechanisms. This is actually stricter than the “black-box” scenario in Mi et al. [20], which assumes the attacker knows the conversion process but not the channel parameters, and more realistic than Zhang et al. [43], which relies on thousands of server queries per identity. In real attacks, these 30 pairs can be obtained by a small number of leaked samples, known identities, or attacker-controlled accounts. It can also be simulated with low-frequency queries to the authentication server.

We observe that despite their claimed algorithmic complexity, the output templates of these systems share a common visual essence: they all preserve high-frequency information while obfuscating low-frequency information. Based on this intuition, the attacker can bypass any system-specific modeling and instead use a generic Gaussian-blur-based high-pass filter as a universal proxy task. We avoided using DCT or DWT to decouple from the methods used in the tested systems. By subtracting a slightly blurred version from the original image and applying simple data augmentations (e.g., varying kernels and strengths), the attacker trains a student model to align this simple high-pass domain with the identity embedding space. The attacker does not need to know any details about the system; instead, the high-pass characteristic is easily observable visually.

The training process is the same as our main text, with simply the known PPFR conversion process replaced with a high-pass

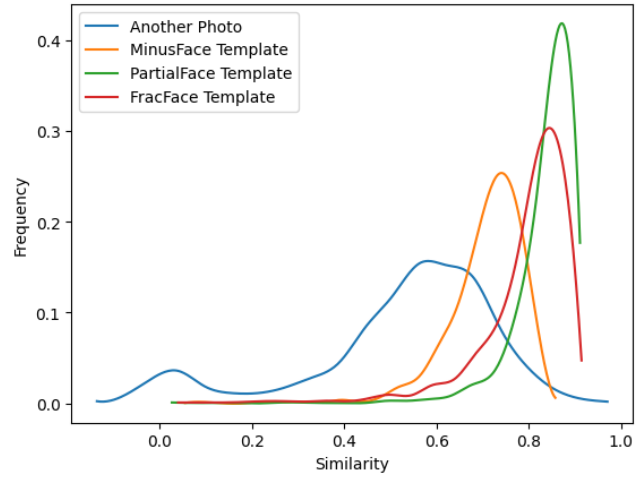


Figure 5: Cosine similarity distributions in ArcFace embedding space. *Source-to-template*: similarity between an image and its own protected template (via the student model). *Cross-image*: similarity between two different images of the same identity. The template-derived embedding is closer to its source image than a second photo of the same person.

filter. During inference, the templates are directly fed into the student model except for MinusFace, for which it is passed through a Gaussian-blur-based high-pass filter to remove low-frequency noise.

We trained only one model to attack all three methods. As shown in Table 7, identity leakage remains strikingly persistent. For all three systems, we achieved over 92% 1-to-1 matching success rate and over 94% re-generation success@5 on Face++ and about 44-57% on the Amazon API. The Amazon API is likely more strict or sensitive to AI-generated images. Notably, the 1-to-1 linkage success rate on LFW remains around 92-96% (Table 7), close to our main experiments.

These results show that, for the frequency-domain PPFR methods evaluated, identity extraction remains feasible even under constrained attacker assumptions. The three systems share a common characteristic: their output representations retain strong coupling with simple high-pass filtering operations, which is what makes this proxy-based attack possible. We note that this particular attack is specific to the frequency-domain family and does not apply to non-frequency methods such as CanFG. However, CanFG remains vulnerable under the oracle-access threat model (Section 5), indicating that identity leakage is not unique to frequency-domain design. Across both settings, the results suggest that as long as templates preserve recognition utility, they tend to retain recoverable identity information, and current frequency-domain defenses do not adequately address this.

7 Similarity Distribution

To quantify identity leakage independent of Arc2Face or commercial APIs, we analyze cosine similarity in ArcFace embedding space between (1) two images of the same person and (2) an image and

Table 6: Comparison with FracFace defensive claims. While FracFace [6] measures protection by frequency channel disruption, we evaluate actual identity leakage. High disruption rates fail to prevent extraction, as FaceLinkGen achieves near-total recovery through commercial-grade verification.

	Venue	Protection Tested in FracFace [6]	Protection Tested Using Our Attack	Protection Tested Using Our Attack (5 trials)
PartialFace [20]	ICCV 2023	0.680	0.002	0.000
MinusFace [21]	CVPR 2024	0.850	0.011	0.004
FracFace [6]	NeurIPS 2025	1.000	0.015	0.008

Table 7: Assumption-constrained attack results. Regeneration Success@5 is reported for Face++ and Amazon API; matching accuracy is the maximum 1-to-1 linkage accuracy at the optimal threshold.

Method	Face++	Amazon API	Matching
FracFace	0.946	0.473	0.949
PartialFace	0.946	0.447	0.925
MinusFace	0.963	0.570	0.962

its protected template (Figure 5). Some near-zero similarities are due to dataset noise [7, 32]; we focus on the main cluster. Across all three methods, the similarity between an image and its template exceeds that between two different images of the same person.

8 Transfer to Attacking De-Identification Systems

Our primary focus is on PPFR systems, which retain machine-readable identity information while rendering the original face unreconstructible from the template by removing some visual information. However, a complementary line of work pursues the opposite objective: preserving visual recognizability to humans while disrupting machine-based face recognition [27, 29, 38]. These systems apply adversarial noise or synthetic makeup so that the protected face remains human-recognizable but causes FR systems to fail.

This design introduces a vulnerability symmetric to the one we identify in PPFR. PPFR systems leak identity by preserving machine-readable mutual information; these perception-consistent de-identification systems leak identity by preserving human-perceivable information. In both cases, sufficient identity signal is retained for extraction. Notably, the same learning-based pipeline suffices to recover this information in both settings. As long as this mutual information can be aligned to a publicly accessible embedding space, a student model can recover it.

We selected TIP-IM [38] as a representative dodging method. We note that many methods [12, 27] conflate dodging with impersonation, which are not equivalent: an FR system can simultaneously associate a face with both the original identity and the impersonation target. In privacy-purposed de-identification scenarios, users care about dodging, not impersonation, which is closer to attacking

Table 8: Identity leakage evaluation on TIP-IM de-identification. We report cosine similarity and linkage accuracy using the standard FR model f and our student model d on protected images x_p .

	TIP-IM		CanFG	
	$f(x_p)$	$d(x_p)$	$f(x_p)$	$d(x_p)$
Cosine Similarity to $f(x)$	-0.081	0.648	0.042	0.520
Linkage Accuracy (same image)	0.000	0.997	0.10	0.997
Linkage Accuracy (cross image)	0.000	0.939	0.04	0.825

an FR system. We evaluate on a small subset of CASIA-WebFace (2,082 images from 408 identities).

We report $\text{sim}(f(x), f(x_p))$, where x is the original image and x_p is the protected image, reflecting how well the de-identification method deceives the original FR model, and $\text{sim}(f(x), d(x_p))$, measuring whether our distillation pipeline can recover the identity information that the protection attempts to suppress. The results in Table 8 show that although TIP-IM provides perfect protection against ArcFace, this is almost completely undermined by our distillation pipeline. This confirms that adversarial perturbation suppresses identity in one model's embedding space while leaving it recoverable by a student aligned to a different space, suggesting a structural limitation of adversarial de-identification as a privacy mechanism.

We additionally evaluate on CanFG [35], which can be viewed as both a non-frequency-domain PPFR method and a non-adversarial de-identification method. As discussed in our threat model, CanFG conflates PPFR with facial de-identification and places its security entirely in the secrecy of the conversion model: an assumption that does not hold when the service provider is the adversary. Its deterministic transformation also leaves it vulnerable to linkage attacks. Under our pipeline, the results are also shown in Table 8, confirming that the same distillation framework used for frequency-domain PPFR and adversarial de-identification generalizes to non-adversarial, non-frequency-based settings as well.

9 Soft Identity Leakage: Beyond Unique Identifiers

Beyond hard identity recognition, the exposure of soft biometric attributes presents a significant privacy risk. Characteristics such

as skin color, age, and gender are sensitive personal data that enable unauthorized profiling and algorithmic discrimination. Privacy frameworks like the Canadian Privacy Act [5] explicitly protect race and age. A robust privacy-preserving system must therefore prevent the recovery of these attributes from its templates.

In the public rebuttal of FracFace [6] on OpenReview [1], the authors claimed successful obfuscation of age, gender, and ethnicity. They cited a human perception study where participants reported less than 13% usage of these biometrics for identity inference, with over 76% of participants relying on guesswork. However, our empirical results in Figure 1 demonstrate that these soft biometrics remain visible in the re-generated faces. This matches with previous research, which indicates that ArcFace embeddings retain such information [18, 23]. Because our extracted embeddings closely resemble the original facial embeddings, we hypothesize that a model can learn a direct mapping from the template to these attributes without facial reconstruction.

To assess this, we appended a multi-head prediction MLP to the distilled student model and trained it for one epoch on the FairFace dataset [17] to infer soft biometrics, predicting age, gender, and race (7 categories). Evaluation was performed on 10k images. As reported in Table 9, for all these protected templates, soft-biometrics can be inferred at nearly the same accuracy as from the original images for FracFace and PartialFace. For MinusFace, the reconstruction is less accurate, though a substantial amount of information is still recovered.

For completeness, it is worth noting that some methods, such as CanFG [35] and FaceAnonymizer [2], intentionally retain soft biometric traits to facilitate auxiliary tasks. We contend that this design decision should be scrutinized more carefully. In PPFR systems, preserving soft biometric details furnishes adversaries with highly sensitive information while offering no advantage for recognition accuracy. These attributes are a focal point of privacy regulations and may necessitate stricter protections [24]. We did not assess soft-biometric leakage on faces protected by CanFG, as such leakage is immediately apparent: as illustrated in Figure 2, CanFG-generated faces, by design, retain nearly all soft biometric characteristics by design.

Table 9: Soft biometric leakage from protected templates. Original-image ArcFace embeddings are included as an upper bound.

Method	Race Acc $\uparrow$	Gender Acc $\uparrow$	Age MAE $\downarrow$
FracFace	0.673	0.925	4.737
MinusFace	0.569	0.893	5.883
PartialFace	0.663	0.932	4.952
Random Baseline	0.146	0.493	21.51
Original Image	0.700	0.949	4.584

10 Future Directions

We suggest several potential pathways for future PPFR designs and evaluations, primarily focusing on stronger defensive mechanisms and broader vulnerability assessments.

Cryptographic and Key-Based Hardening. One rigorous approach is to incorporate secret keys into the conversion process, similar to Yuan et al. [41]. This serves as a multi-factor authentication system (requiring both biometrics and a key), preventing attackers (including our method) from converting a face without the secret. Alternatively, systems may revert to formal cryptographic methods like [3]. While traditionally viewed as computationally expensive, modern resources make this trade-off acceptable; for instance, Jindal et al. [15] reports only 2.83ms processing time per face pair. Importantly, these computational costs can act as an effective client-side constraint against brute-force attacks (conceptually similar to slow hashing), enhancing privacy while remaining imperceptible to regular users. Again, we want to emphasize that de-ID or reversible face encryption methods *cannot* be used for PPFR tasks, as they either make the face completely useless for recognition or the reversibility compromises the privacy-preserving nature.

De-identification: Fooling Human Perception Instead of Machines. Current de-identification methods aim to fool FR systems while keeping the identity visually intact to human observers. However, this is almost guaranteed to be attacked due to the information still being intact. However, research suggests that human and machine perception of facial similarity diverges [26]. We propose an alternative direction: rather than hiding identity from machines while preserving it for humans, one could actively remove identity information from the image while optimizing a human perceptual loss to make the result appear identity-consistent to human observers. This inverts the typical de-identification objective and may offer stronger privacy guarantees if real identity information can be completely removed. One possible approach is to reconstruct the face from identity-agnostic geometric cues such as depth maps or Canny edge maps, which preserve appearance structure without encoding biometric identity. Alternatively, we can adopt an inverted approach similar to these frequency-based methods, where we retain only the low-frequency components that are important for human perception and discard the high-frequency components that carry identity information. This mapping could also be N-to-1 such that a human can still *recognize* the face, but neither a human nor a machine can re-identify the face due to lost information.

11 Conclusion

This paper shows that pixel-level reconstruction metrics may not adequately capture identity-level leakage in the frequency-domain PPFR systems we evaluated. Identity information can be extracted from protected templates via a straightforward distillation pipeline and used for linkage and face regeneration without recovering original pixels. FaceLinkGen achieves over 98.5% matching accuracy and above 96% regeneration success on three frequency-domain PPFR systems, suggesting that these methods do not provide the level of identity privacy their pixel-level evaluation results imply.

We hope these findings encourage the adoption of identity-centric evaluation alongside existing metrics in frequency-domain PPFR research, and motivate exploration of stronger protection mechanisms, such as cryptographic approaches, that offer more formal privacy guarantees.

References

- [illegible]

- [41] Zhuowen Yuan, Zhengxin You, Sheng Li, Zhenxing Qian, Xinpeng Zhang, and Alex Kot. 2022. On Generating Identifiable Virtual Faces. In *Proceedings of the 30th ACM International Conference on Multimedia*. ACM, Lisboa Portugal, 1465–1473. doi:10.1145/3503161.3548110
- [42] Kai Yue, Richeng Jin, Chau-Wai Wong, Dror Baron, and Huaiyu Dai. 2023. Gradient Obfuscation Gives a False Sense of Security in Federated Learning. In *32nd USENIX Security Symposium (USENIX Security 23)*. USENIX Association, Anaheim, CA, 6381–6398. <https://www.usenix.org/conference/usenixsecurity23/presentation/yue>
- [43] Hui Zhang, Xingbo Dong, YenLung Lai, Ying Zhou, Xiaoyan Zhang, Xingguo Lv, Zhe Jin, and Xuejun Li. 2024. Validating Privacy-Preserving Face Recognition Under a Minimum Assumption. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 12205–12214. doi:10.1109/CVPR52733.2024.01160